

Customer Churn: Who to Save, Why They Leave, What It Is Worth

Business report | IBM Telco dataset, 7,043 customers | model: XGBoost

Executive summary

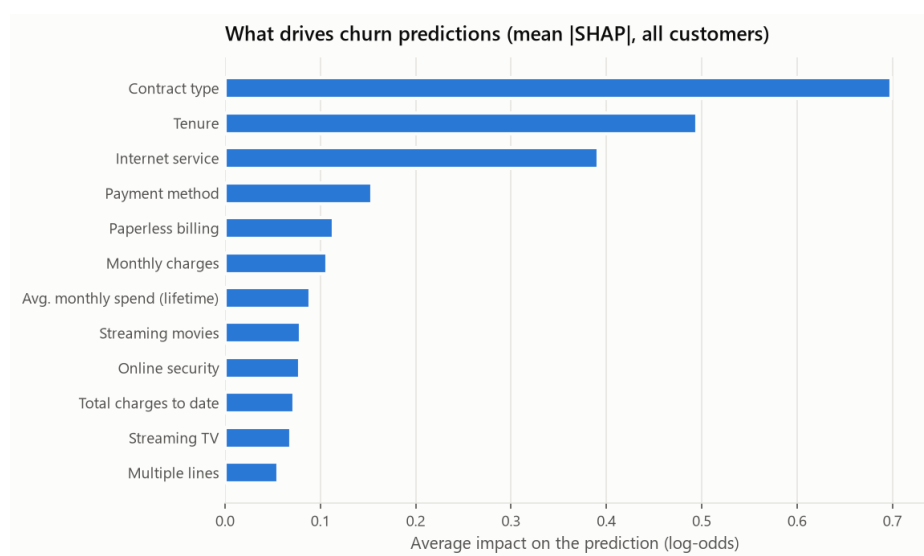
- **Churn is expensive:** 26.5% of customers left last month, taking 30.5% of monthly revenue with them. Across the 5,174 remaining customers, an estimated **\$820,439** of annual revenue is at risk.
- **The risk is concentrated:** 1,145 active customers (22%) carry **63%** of the expected margin loss. They are the retention target list.
- **Why they leave:** month-to-month contracts (43% churn vs. 3% on two-year contracts), short tenure (20% of all churners leave in their first month), fiber internet and missing security/support add-ons.
- **Decide with money, not accuracy:** contacting every customer above a churn probability of **0.32** instead of the default 0.50 reaches 74% of churners instead of 52% and earns **\$2,054 (+19%)** more net value on held-out customers. Targeting by each customer's expected value adds more (+31% vs. 0.50).
- **Recommendation:** run a retention campaign on the 1,145 customers, led by annual-contract offers and early-life onboarding; expected net value **\$23,696** over 12 months at a cost of \$22,900. Launch it with a control group to measure the real save rate.

Key numbers

Measure	Value	Measure	Value
Customers	7,043	Active customers	5,174
Churn rate (last month)	26.5%	Recommended for contact	1,145
Annual revenue at risk	\$820,439	Expected campaign net value	\$23,696
Model ROC-AUC (held out)	0.849	Money-optimal threshold	0.32

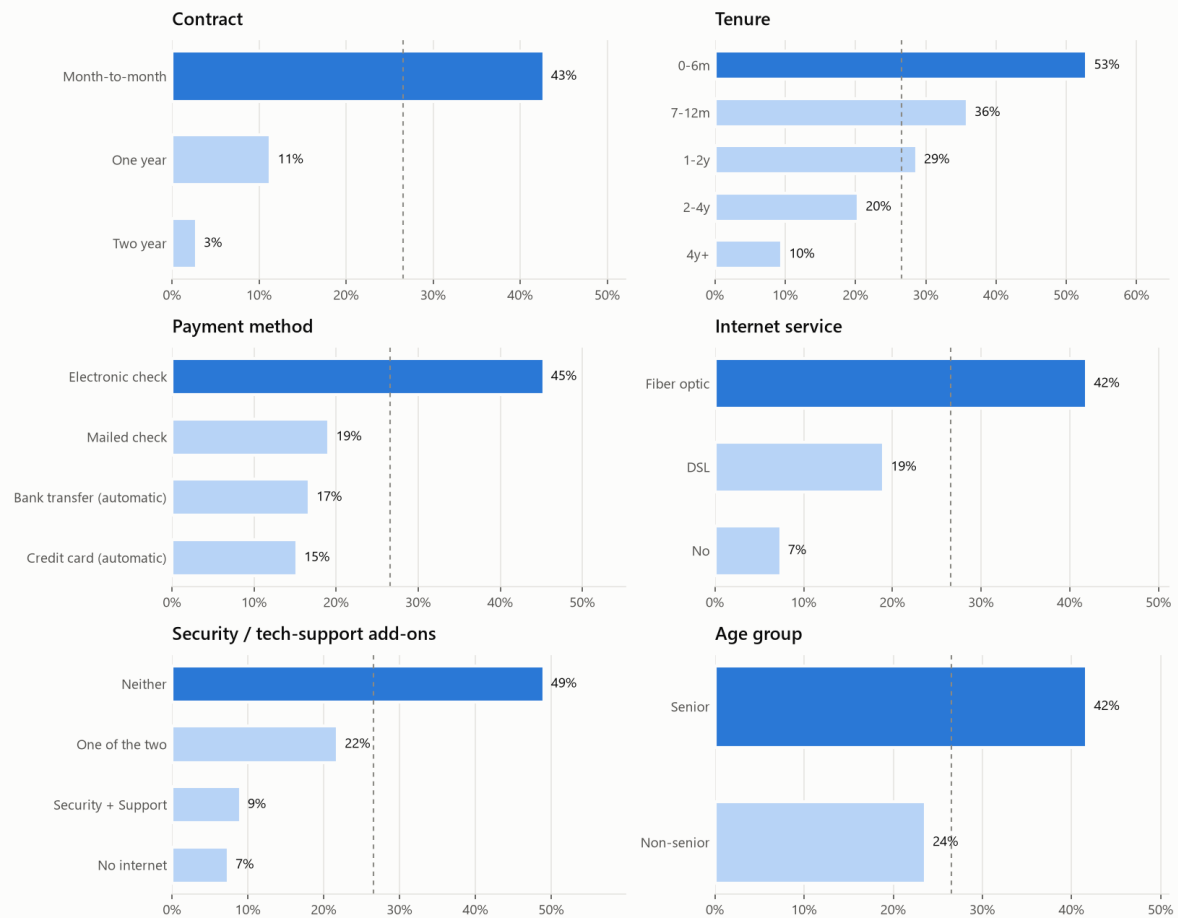
Assumptions (adjustable in the dashboard): a retention offer costs \$20 per contacted customer; 30% of contacted churners are saved; a saved customer is worth 12 months of their bill at a 30% margin.

What drives churn

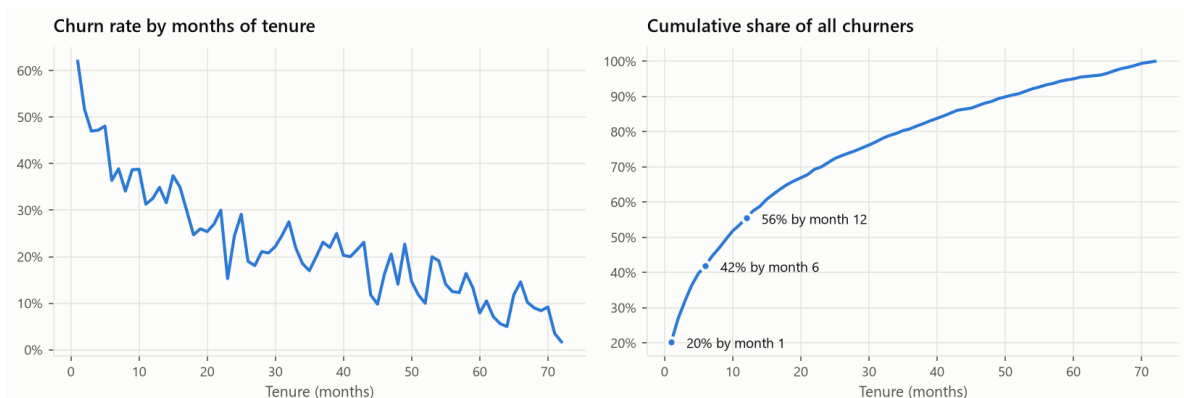


1. Who is leaving and why

Churn rate by customer segment (dotted line = all customers, 26.5%)



The riskiest combination is **Month-to-month + Fiber optic**: 55% churn across 2,128 customers. Customers with both online security and tech support churn at a fraction of the rate of those with neither, and electronic-check payers churn about three times as often as customers on automatic payment.

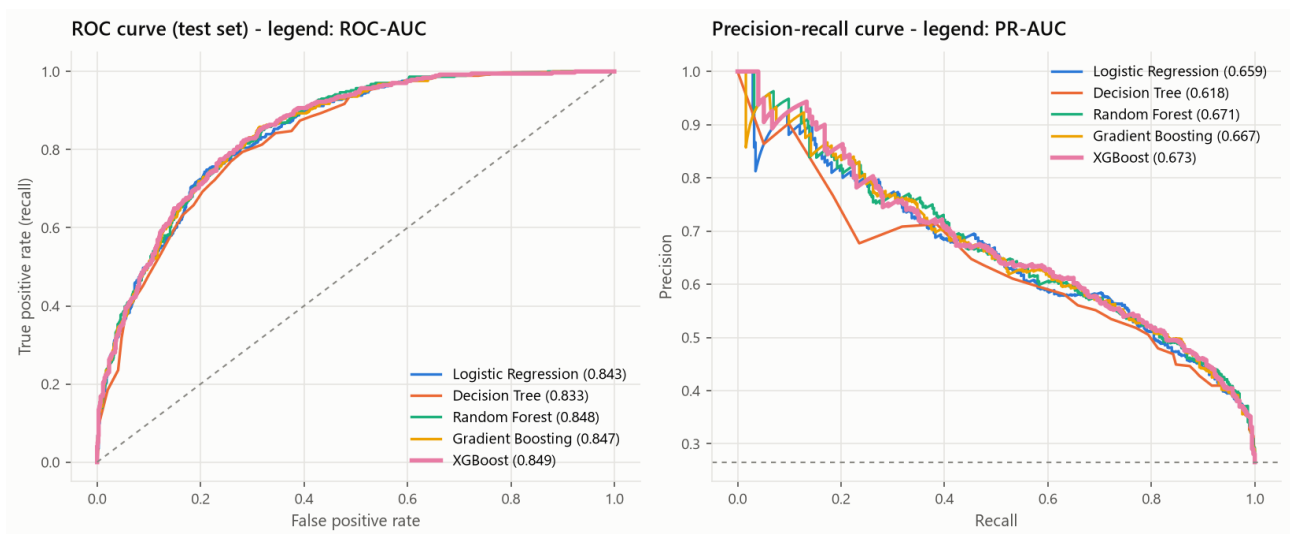


Churn is front-loaded: 42% of all churners leave within six months and 56% within a year. The first months are where retention is won or lost.

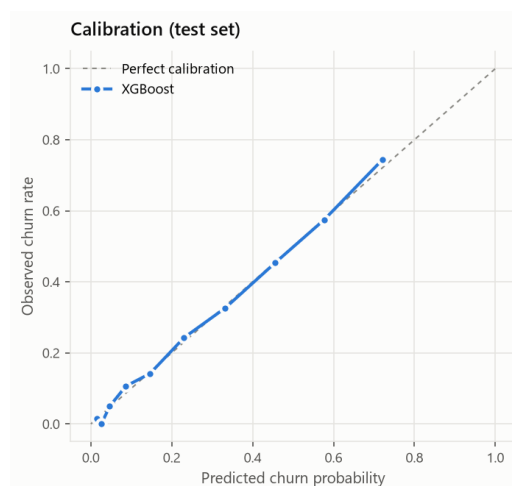
2. How reliable is the prediction?

Five model families were tuned and compared with 5-fold cross-validation on 80% of customers; the remaining 20% were held back and scored once, at the end. The top four models perform almost identically; the champion was chosen in advance on cross-validated ROC-AUC.

Model	CV ROC-AUC	Held-out ROC-AUC	PR-AUC	Recall @0.5	Recall @t*	Net value @t*
Logistic Regression	0.8448	0.843	0.659	52%	77%	\$12,992
Decision Tree	0.8313	0.833	0.618	45%	79%	\$12,532
Random Forest	0.8461	0.848	0.671	50%	73%	\$12,763
Gradient Boosting	0.8480	0.847	0.667	51%	73%	\$12,777
XGBoost	0.8485	0.849	0.673	52%	74%	\$12,736

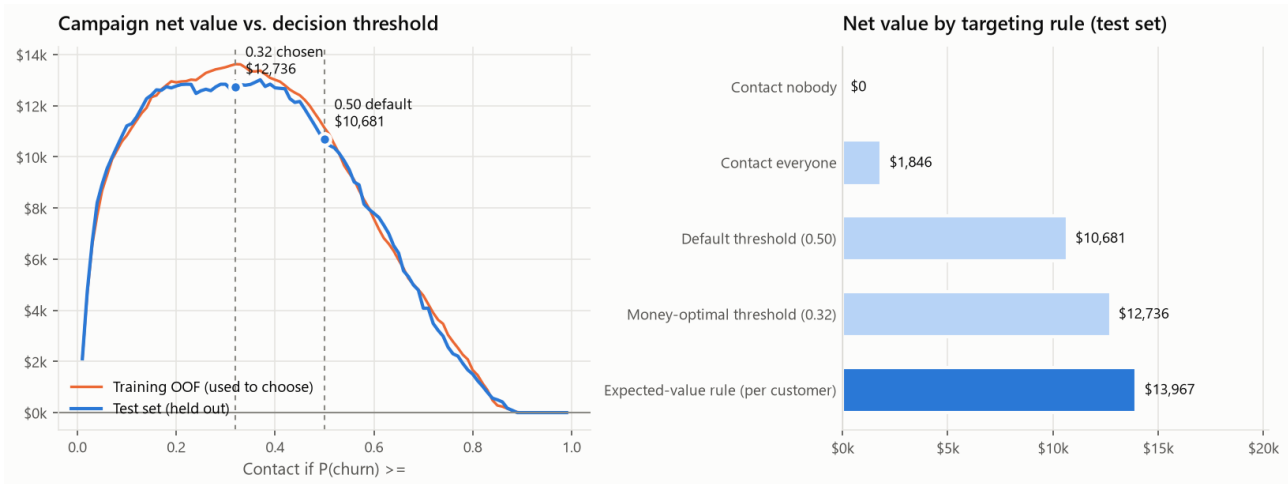


All models except the single decision tree rank customers almost identically. The top 10% of scores contain 2.8x the average churn rate.



The predicted probabilities are well calibrated (a predicted 40% really means about 40%), which is what makes expected-loss and campaign-value figures trustworthy.

3. Choosing who to contact: a money decision



Results on 374 churners among the 1,409 held-out customers:

Targeting rule	Contacted	Churners reached	Cost	Net value
Contact nobody	0	0	\$0	\$0
Contact everyone	1,409	374	\$28,180	\$1,846
Default threshold (0.50)	305	194	\$6,100	\$10,681
Money-optimal threshold (0.32)	508	278	\$10,160	\$12,736
Expected-value rule (per customer)	535	277	\$10,700	\$13,967

A missed churner costs far more than a wasted offer, so the best threshold sits well below 0.5. The value curve is flat between roughly 0.2 and 0.45: the decision is robust to the exact cut-off, but not to using the default.

4. Who to call first

Active customers ranked by expected loss (churn probability x customer value), each with the reasons behind the score (SHAP) and the recommended action:

Customer	P(churn)	Exp. loss	Main reasons	Recommended action
1393-IMKZG	86%	\$296	Short tenure (only 1 month) Month-to-month contract Fiber-optic internet plan	Early-life onboarding call: check the setup, walk through the bill Offer a discounted 12-month contract
2081-VEYEH	76%	\$294	Short tenure (only 3 months) Month-to-month contract Fiber-optic internet plan	Early-life onboarding call: check the setup, walk through the bill Offer a discounted 12-month contract
1941-HOSAM	87%	\$282	Short tenure (only 1 month) Month-to-month contract Fiber-optic internet plan	Early-life onboarding call: check the setup, walk through the bill Offer a discounted 12-month contract
7145-FEJWU	74%	\$281	Month-to-month contract Fiber-optic internet plan Lifetime average bill of \$106.30/month	Offer a discounted 12-month contract Proactive fiber service check (speed test, outage history)
5150-ITWWB	82%	\$280	Short tenure (only 3 months) Month-to-month contract Fiber-optic internet plan	Early-life onboarding call: check the setup, walk through the bill Offer a discounted 12-month contract
5760-IFJOZ	72%	\$279	Short tenure (only 3 months) Month-to-month contract Fiber-optic internet plan	Early-life onboarding call: check the setup, walk through the bill Offer a discounted 12-month contract
7577-SWIFR	87%	\$278	Short tenure (only 1 month) Month-to-month contract Fiber-optic internet plan	Early-life onboarding call: check the setup, walk through the bill Offer a discounted 12-month contract
3292-PBZEJ	69%	\$276	Month-to-month contract Fiber-optic internet plan Pays by electronic check	Offer a discounted 12-month contract Proactive fiber service check (speed test, outage history)
6734-GMPVK	73%	\$276	Short tenure (only 5 months) Month-to-month contract Fiber-optic internet plan	Early-life onboarding call: check the setup, walk through the bill Offer a discounted 12-month contract
4132-KALRO	74%	\$269	Short tenure (only 4 months) Month-to-month contract Fiber-optic internet plan	Early-life onboarding call: check the setup, walk through the bill Offer a discounted 12-month contract
4273-MBHYA	83%	\$268	Short tenure (only 1 month) Month-to-month contract Fiber-optic internet plan	Early-life onboarding call: check the setup, walk through the bill Offer a discounted 12-month contract
6350-XFYGW	78%	\$265	Short tenure (only 4 months) Month-to-month contract Fiber-optic internet plan	Early-life onboarding call: check the setup, walk through the bill Offer a discounted 12-month contract

Behavioural segments

Segment	Customers	Churn rate	Avg tenure	Avg bill	Exp. loss (active)
New, high-spend	2,410	47%	15 mo	\$77	\$128,122
New, low-spend	1,509	24%	10 mo	\$29	\$24,916
Loyal, high-spend	2,152	16%	57 mo	\$91	\$86,749
Loyal, low-spend	972	4%	54 mo	\$30	\$6,344

The **New, high-spend** segment holds 52% of the expected loss among active customers.

5. Recommendations

- **Launch the campaign** on the 1,145 customers above the 0.32 threshold (or use the per-customer expected-value rule), working the list in expected-loss order.
- **Convert month-to-month customers** to discounted 12-month contracts - contract type is the strongest single driver of churn.
- **Invest in the first 90 days:** onboarding calls, setup checks and a walk-through of the first bill, because churn is concentrated in the first months.
- **Bundle security and tech support** (e.g. a free three-month trial) for fiber customers who have neither.
- **Move electronic-check payers to automatic payment** with a small bill credit.
- **Measure, then re-tune:** hold back a random control group so the real save rate replaces the assumed one, then re-optimize the threshold in the dashboard.

Limitations

The IBM Telco data is a single snapshot without support-call, usage-history or payment-delay columns, so those drivers could not be measured. Campaign economics rest on assumptions (offer cost, save rate, margin) that should be replaced with measured values. Scores for existing customers are out-of-fold: every customer was scored by a model that did not see them during training.